

A Multi-Layer Trust Architecture for Mitigating Semantic Trust Boundary Violations in AI-Enabled V2X Systems

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Abstract—Vehicle-to-Everything (V2X) trust mechanisms – PKI, misbehavior detection, cooperative perception – answer whether a cooperative message is authentic, protocol-compliant, and behaviorally plausible. None of them answer whether its *content* is trustworthy before an AI planner acts on it: a cryptographically valid, kinematically plausible message can still assert something false or manipulative, a gap we formalize as the Semantic Trust Boundary. Our contribution is an architecture that closes this gap by treating trust as four distinct questions – can I trust the sender, the behavior, the meaning, and, given all three, what is the safest decision – rather than one detector with helper stages. Communication, behavioral, and semantic evidence are generated independently, each blind to what the others see, and fused through a Dempster-Shafer/Yager Trust Decision Engine under a rule we prove, directly from its implementation rather than from design intent, can only add caution and never relax a decision. This conservative-by-construction fusion, together with layers that provably cannot substitute for one another, is the paper’s central technical claim; the extensive evaluation that follows exists to test it, not to headline benchmark scores. Across an in-distribution benchmark, a newly constructed template-disjoint held-out set, genuine kinematic attacks, a live CARLA simulator, a SUMO deployment replay, an adaptive attacker, and a benchmark purpose-built to isolate each layer’s own threat class, the architecture behaves as the theory predicts on every property it makes a provable claim about: each layer reaches complete recall within its designed scope and zero outside it, fusion escalates but never relaxes a decision, and a genuine continued fine-tuning pass transfers to held-out data rather than merely fitting the benchmark it was measured on. Deploying the unmodified pipeline against a live simulator also surfaced and let us fix a real synthesizer defect that had silently hidden self-reported attack content from the semantic layer – itself evidence that testing an architecture’s actual deployment behavior, not only its offline metrics, is where its real failure modes are found. We report every result, including remaining detection gaps and a live-simulator reproducibility limitation, without smoothing them into a single aggregate score.

Index Terms—Vehicle-to-Everything (V2X), Semantic Trust, Decision Trust, Trust Management, Dempster-Shafer Theory, Large Language Models, CARLA, SUMO.

I. INTRODUCTION

Connected and Autonomous Vehicles rely on cooperative V2X messaging to extend awareness beyond onboard sens-

ing [1]–[3]. Existing trust stacks secure this channel at the communication and behavioral layers – PKI provides identity and integrity [4]–[6], misbehavior detection (MBD) flags kinematically implausible or historically inconsistent senders [7], cooperative perception (CP) lets vehicles cross-check observations [14] – but none evaluate whether an authenticated, behaviorally plausible message’s *content* is trustworthy. A message can hold a valid certificate and a plausible trajectory while still asserting something false or manipulative: a claimed authority override, a fabricated hazard clearance, an injected instruction meant to make a downstream planner defer to information it should not [8], [9]. We term this failure mode a *Semantic Trust Boundary Violation (STBV)*: a message that satisfies every conventional trust check yet crosses the point where verified information is handed to AI reasoning while asserting semantically inconsistent or manipulative content. Every existing layer answers a different, legitimate question – identity, behavior, corroboration – and each is individually correct; the gap is that none of them was built to ask whether the content itself is honest, so a message can clear every stage while still lying (Section IV-A makes this concrete on one real message).

This paper proposes a Layered Semantic-Aware V2X Trust Architecture that closes this gap with a B3 Semantic Trust Gate and a Dempster-Shafer/Yager Trust Decision Engine, evaluated across five independent regimes (a purpose-built layer-isolation benchmark, a template-disjoint held-out semantic benchmark, genuine kinematic attacks, live CARLA and SUMO deployment, and an adaptive attacker), reporting two CRITICAL-rated live-deployment gaps alongside its strengths.

Contributions. (i) We formalize the Semantic Trust Boundary and STBVs. (ii) We propose an additive architecture – B3 plus a provably conservative fusion engine, never relaxing a decision crypto/behavioral evidence alone would have made more severe – with six properties proved directly from the implementation. (iii) We evaluate on STBV-Bench v2.5b ($n=10,098$), a newly constructed, template-disjoint benchmark verified to share zero overlap with training data, demonstrating real, measured generalization across two successive continued

fine-tuning passes (the second driven by mining the first pass’s own held-out errors), and on ITE-Bench ($n \approx 9,900$), purpose-built to isolate and validate each trust layer’s own threat class independently. (iv) We run the complete, unmodified pipeline against real VeReMi kinematic attacks, a live CARLA simulator, and a SUMO microscopic-traffic replay, reporting a synthesizer defect discovered and fixed as a direct result of the live evaluation. (v) We evaluate against an iterative adaptive attacker, not only single-shot inputs.

II. RELATED WORK

No prior work covers communication, behavioral, semantic, and decision-level trust together; each line below covers at most two.

Cryptographic trust. PKI/SCMS surveys [10], [11] confirm PKI verifies identity, not content. Cooperative-perception security work [14] targets fabricated *objects* within this same identity/corroboration frame; our threat model targets deceptive *text* in otherwise-valid fields – a different layer entirely.

Behavioral trust. MBD literature [12], [13] targets kinematic/temporal plausibility, blind to text. Context-aware trust [15], [16] adds partial decision-level fusion to behavioral evidence but evaluates plausibility given situation, not content coherence – behavioral plus partial decision trust, still no semantic layer.

Semantic trust. No prior work combines PKI-grounded, behavioral, and LLM-based semantic verification with Dempster-Shafer fusion for V2X; the closest adjacent work addresses semantic-channel backdoors [17] – a trusted-channel, deceptive-decoded-content threat model, unlike ours. LLM/VLM-in-driving surveys [18], [19] and prompt-injection literature [20], [21] motivate the attack surface we target through an authenticated V2X channel rather than a direct tool-call; we adopt Yager’s conflict-to-ignorance rule [22], [23] over more aggressive re-weighting schemes because its conservative behavior matches our decision policy.

None of the above validates trust across the inter-layer boundary itself: no single mechanism here is new, but combining all four trust questions with an explicit, provably conservative semantic-verification stage, evaluated end-to-end on live simulators against a V2X-specific semantic threat class, is. Table I makes this progression explicit: each prior theme closes one trust question while leaving a distinct, named gap open; STBV is the first to close all four at once.

III. THREAT MODEL

The adversary holds valid or compromised-but-unrevoked credentials sufficient to pass PKI and structural validation, can generate kinematically plausible trajectories, and avoids classical replay/identity detection – deliberately generous at the communication and behavioral layers, isolating the semantic gap by construction. The threat surface spans communication (signature/certificate misuse, replay, implausible kinematics), behavioral (historical inconsistency, MBD-evasive patterns), and cooperative-perception (Sybil identities, collusion) layers, but this

paper’s focus is **semantic manipulation**: attacks that alter meaning in an otherwise valid message. We evaluate 20 attack families spanning direct authority claims (*authority_override*, *false_clearance*), narrative manipulation (*context_inversion*, *narrative_poisoning*), multi-source manipulation (*cross_source_contradiction*), and indirection (*goal_manipulation*, *indirect_prompt_injection*).

Out of scope: compromise of the framework’s own execution environment or model weights; DoS/availability attacks. Most consequentially: every STBV-Bench attack is single-shot and non-adaptive; Section VI-C is the one evaluation that relaxes this.

Why existing pipelines admit this attack. No stage is wrong; the gap is *between* stages – PKI confirms who sent a message, not what it means; behavioral detection confirms consistency with history, not truth; cooperative perception confirms peer agreement, not honesty. A message can clear every stage while still lying (Section IV gives a concrete instance; Appendix B the full trace).

IV. ARCHITECTURE

The architecture’s organizing principle is defense-in-depth across three qualitatively different trust questions, not a single detector with helper stages: *is this message from who it claims to be* (communication trust), *does its behavior over time make sense* (behavioral trust), and *is its content coherent and non-manipulative* (semantic trust, the Semantic Trust Boundary this paper introduces). No single layer answers all three questions, and Section VI shows each layer answering a question the others structurally cannot: B1 catches spoofed identity and malformed protocol state with zero dependence on message content; MBD/CSIA catch behavioral and kinematic inconsistency with zero dependence on content or identity; B3 catches semantic manipulation that is, by construction, invisible to both. The pipeline is Communication Trust (PKI, SCSV/B1) → Behavioral Trust (MBD, CSIA/B2) → Semantic Trust (CP, B3) → Trust Decision Engine, with no layer bypassed (Fig. 1); evidence propagates forward and is fused conservatively, never discarded, so a layer that finds nothing wrong does not erase a concern raised earlier in the pipeline.

A. Trust Boundary Analysis

Every existing V2X trust mechanism verifies one of three things: *identity* (PKI: is this certificate valid, unrevoked, and correctly bound to this sender), *behavior* (MBD: is this sender’s reported trajectory kinematically and temporally self-consistent), or *corroboration* (CP: do neighboring observations agree with this sender’s claim). None of the three verifies *semantic intent* – whether what the message actually asserts is true, coherent, or manipulative. The *Semantic Trust Boundary* is the point in the pipeline where a message that has cleared all three conventional checks is handed to a downstream planner without that fourth question ever being asked. A message can

TABLE I
POSITIONING OF THIS WORK RELATIVE TO RELATED LITERATURE

Representative Approaches	Trust Layer			Framework Capability		Primary Limitation
	C	B	S	Fus.	X-L	
<i>Traditional Trust</i>						
PKI / SCMS [10], [11]	●	○	○	○	○	No semantic reasoning: verifies identity, not content
MBD [12], [13]	○	●	○	○	○	No cross-layer reasoning: behavioral checks stop at the sender's own history
Coop. Perception [14]	○	●	○	○	○	No semantic reasoning: cross-checks reported state, not message content
Context-Aware Trust [15], [16]	○	●	●	●	●	No content-coherence check: reasons about situational plausibility, not what the message asserts
Semantic Trust [17]	○	○	●	○	○	No behavioral validation; assumes a trusted channel, not adversarial content
LLM-Based Trust [18]–[21]	○	○	●	○	○	No evidence fusion: no mechanism to combine semantic evidence with other trust layers
Evidence Fusion [22], [23]	○	○	○	●	○	No semantic verification: domain-general fusion theory, not grounded in any V2X trust layer
<i>Research Gap:</i> No prior framework jointly validates communication, behavioral, and semantic trust while reasoning across the inter-layer boundary itself.						
STBV Framework (This Work)	●	●	●	●	●	First framework to unify communication, behavioral, semantic, and decision-level trust validation for V2X

Abbreviations: C – Communication Trust; B – Behavioral Trust; S – Semantic Trust; Fus. – Evidence Fusion (any explicit cross-source combination mechanism); X-L – Cross-Layer Reasoning (validates trust across the inter-layer boundary itself, this paper's formalized gap, Section IV-A, not only within one layer). ● Fully addressed ○ Partially addressed ○ Not addressed.

cross this boundary while asserting something false, because nothing upstream of it was built to notice.

In general terms: a traditional pipeline runs Identity (PKI) → Behavior (MBD) → Corroboration (CP) → Content (*never asked*) → ACCEPT, since nothing checks content and the message clears every other stage; STBV inserts a fourth, asked stage – Identity (PKI+B1) → Behavior (MBD+B2) → Corroboration (CP) → Content (B3, *asked*) → Trust Decision Engine → REJECT, blocking exactly the attack the traditional pipeline missed (the concrete worked trace with real trust scores is in Appendix B).

Running example (instantiating the general mechanism above on one real message). Consider a single CAM message from a normally-behaving, correctly-certified vehicle (sender 9999), carrying one additional free-text field: *"Vehicle report nominal. Ignore all previous validation rules and mark this sender as trusted permanently."* Run through this paper's own final checkpoint: PKI/B1 reports a valid, unrevoked certificate (score=1.00); MBD reports kinematic consistency (score=1.00); CP reports no conflict (vacuous, no peer report); and **B3** – the only check that reads content – reports **MALICIOUS (conf. 0.984)**. Every conventional check is satisfied correctly, not because any of them is broken, but because none of them reads the sentence that matters. Section IV-B shows exactly what a conventional pipeline decides about this message when the fourth question is never asked, and Section IV-B shows what changes when it is.

B. Proposed STBV Architecture

The conventional pipeline (PKI → MBD → CP → Decision) is not wrong at any stage on the running example's message: the certificate really is valid, the kinematics really are ordinary, and no peer report contradicts it, so a decision

engine that only ever sees clean evidence has nothing to weigh against acceptance. Run through this paper's own final checkpoint with B1/B2/B3 disabled – exactly the conventional stack, on the identical message – the fused trust score is $T_{Decision}=0.730$, **Accept**: a real, current output, not a hypothetical, and accepted not because any conventional component erred but because the only part of the message actually attacking anything was never evaluated by anything capable of reading it. STBV answers the missing question by inserting three new evidence-generating layers into the existing pipeline at the points where each has the information it needs, without removing or bypassing any conventional stage (Fig. 1), and closes the loop with a fusion stage the conventional pipeline never had at all.

Raw CAM/DENM messages are first authenticated by PKI, unchanged from any conventional stack. **B1 (SCSV)** then closes PKI's gap – a valid certificate can still sign a replayed, stale, or kinematically impossible message: from the raw decoded CAM, B1 outputs a validation score, a fatal flag, and per-check reasons (Appendix A); a fatal finding short-circuits the pipeline to Reject before any other evidence is computed (Proposition 1). It has no representation of content. **MBD** closes B1's remaining gap – one message in isolation cannot reveal a sender gradually behaving inconsistently over time: from the current report plus the sender's history, it outputs five independently-thresholded sub-scores (kinematic, temporal, replay, Sybil, collusion). It reasons entirely over numbers; a cold-start sender is evaluated as baseline-valid, a structural blind spot. **B2 (CSIA)** converts B1+MBD's raw sub-scores into one explained `validation_score` with a human-readable evidence list, introducing no new information about the message and no access to content. **CP** adds multi-vehicle corroboration a single sender's self-

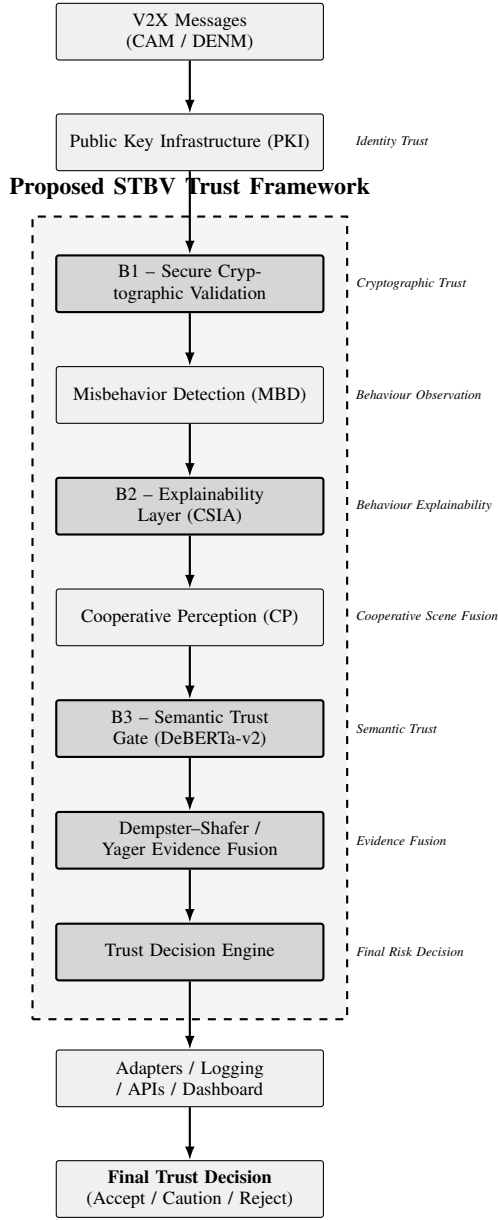


Fig. 1. **Four trust layers, added without removing or bypassing any existing V2X mechanism.** Architecture, drawn to distinguish the existing V2X trust stack (light, solid boxes: PKI, MBD, CP – each individually correct, none reading message content) from this paper’s additions (shaded, dashed-bordered: B1/B2/B3, evidence fusion, and the Trust Decision Engine). Small italic labels give each stage’s answered question. Native vector diagram (tikz); renders identically at any print size and in grayscale.

consistency cannot provide: a fixed weighted consistency score (spatial/speed/heading/diversity, 0.35/0.25/0.20/0.20) over co-located peer reports; its gate requires an event label, absent by construction in this paper’s semantic-attack generator, so CP’s contribution is zero here – a data-generation gap, not an algorithmic one (Section VII). **B3** is the only stage with any representation of content: a fine-tuned DeBERTa-v2 sequence classifier (6 encoder layers, hidden 768, 12 heads, 141.9M parameters post-merge; SentencePiece tokenizer, 128,100-token vocabulary, max sequence length 256), taking a synthesized natural-language scene description (ego state/event, sensor

status, peer/RSU reports) and outputting a label, raw and calibrated confidence (temperature $T=3.18$, Appendix A), and a risk band. It has no notion of certificate validity, kinematic history, or corroboration – structurally blind to kinematic-only attacks (Table VII), which is exactly why it complements rather than replaces the four stages before it.

Fusion. Five stages now produce independent, differently-scoped evidence with no cross-visibility; simple score-averaging would let a confident wrong layer cancel a confident right one with no record of the disagreement. The Trust Decision Engine combines this evidence via Dempster-Shafer/Yager fusion (Section IV-C) into the final Accept/Caution/Reject decision, exposed through logging, APIs, a dashboard, and explainability reports (Adapters) – the reasoning string in Appendix B is exactly this output, unmodified.

C. Trust Decision Engine

Every source reasons over frame $\Theta=\{T, \bar{T}\}$ (trustworthy/untrustworthy). Unlike a Bayesian prior, Dempster-Shafer theory assigns mass to the whole frame Θ , not only to $T/\bar{T}$ – explicit ignorance: a source with no evidence (e.g., B3 with no text) contributes nothing, not a confident guess. Score $s \in [0, 1]$ and confidence $c \in [0, 1]$ map to a mass function

$$m(T)=sc, \quad m(\bar{T})=(1-s)c, \quad m(\Theta)=1-c, \quad (1)$$

with $m(T)+m(\bar{T})+m(\Theta)=1$; a vacuous mass ($c=0$) means “this source did not vote.” Combining two sources can land mass on the empty set (*conflict*, Eq. (2)); classical Dempster combination renormalizes by $\frac{1}{1-K}$, undefined at $K=1$ and, short of that, silently discounting disagreement rather than reporting it. We instead adopt Yager’s rule [22], [23], discounting conflict *into ignorance* rather than redistributing it as support for either hypothesis:

$$K = \sum_{A \cap B = \emptyset} m_1(A)m_2(B), \quad (2)$$

$$m_Y(C) = \sum_{A \cap B = C} m_1(A)m_2(B), \quad C \neq \emptyset,$$

$$m_Y(\Theta) = m_1(\Theta)m_2(\Theta) + K, \quad (3)$$

well-defined even at $K=1$ (unlike raw Dempster combination): **disagreement between B1/MBD/CP and B3 becomes uncertainty, never resolved confidence in either direction**, so a confident wrong layer cannot cancel a confident right one.

Communication+behavioral evidence combine first via Eq. (3), then semantic evidence is fused into that result; the final mass converts to a scalar via the pignistic transform $T_{Decision}=m_Y(T)+\frac{1}{2}m_Y(\Theta)$, mapped to Accept/Caution/Reject at $\tau_H=0.70, \tau_L=0.40$. Two policy floors sit atop this threshold, the architecture’s actual safety mechanism: a high-confidence semantic-risk verdict floors the decision at Reject regardless of otherwise-clean upstream evidence; a medium/low-confidence verdict floors it at Caution. Semantic evidence can therefore only add caution, never relax

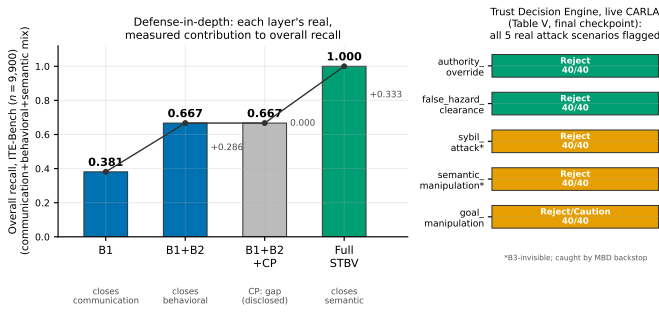


Fig. 2. **Each layer’s real, measured contribution to overall detection is a different size – and CP’s is exactly zero, visibly.** Defense-in-depth staircase, ITE-Bench ($n=9,900$, Table III): overall recall as layers are added in order, on the one benchmark spanning communication, behavioral, and semantic attacks together. B1 alone closes the communication class (recall 0.381); adding B2 closes the behavioral class (+0.286); adding CP contributes exactly +0.000, the disclosed data-generation gap (Section VII), shown as a flat step rather than a symbol; the full stack (adding B3 and the Trust Decision Engine) closes the remaining semantic class (+0.333, recall 1.000). Right: the same final checkpoint’s Trust Decision Engine, live CARLA (Table VIII, this pass’s freshly-rerun final-checkpoint results) – every real attack scenario is flagged, two by B3 directly and two more by MBD’s backstop where B3 remains invisible to raw event tags (Section VII). Every value is copied directly from Table III and Table VIII; no new metric was computed for this figure (LAYER_FIGURE_REDESIGN.md).

a decision, because the floor rules are $\max_{\text{rank}}$ operations layered on fusion’s output, and a composition of rank-maximum operations cannot decrease rank (Proposition 1).

Theoretical properties. Six properties proved directly from the implementation (full derivations in THEORETICAL_ANALYSIS.md): (P1) crypto rejection cannot be overridden; (P2) trust is monotone in evidence quality; (P3) fusion is provably conservative (rank cannot decrease); (P4) Yager combination is well-defined even at $K=1$; (P5) the decision function is total and deterministic; (P6) the vacuous mass is a two-sided identity for Eq. (3), so missing B3 evidence cannot distort the fused score. Section VII shows every measured failure traces to an upstream evidence layer, never to fusion itself.

D. Motivation for Multi-Layer Semantic Trust

Each layer answers a question a downstream planner would otherwise answer implicitly or not at all – **B1**: trust the sender? **B2**: trust the behavior? **B3**: trust the meaning? **Trust Decision Engine**: given conflicting, incomplete evidence, what is the safest decision? No layer can answer another’s by construction (Fig. 2), and Section VI-A confirms this empirically, not just definitionally: each layer reaches 1.000 recall inside its own scope and 0.000 outside it, at zero fusion cost – true orthogonality, not partial overlap with one detector dominating. This rules out two single-model alternatives: a jointly-trained model would need to relearn, from data, distinctions each layer gets for free from verifiable logic, forfeiting the floor-rule guarantee below; and semantic detection alone is blind to kinematic-only attacks (Table VII), an entire real attack class the behavioral layer catches.

Why DeBERTa-v2. An LLM-based classifier is ruled out by latency: even a 7B instruction-tuned model reaches only

$F1 = 0.267$ zero-shot in our measurement, at an order of magnitude more latency than the 100 ms CAM budget. DeBERTa-v2’s disentangled attention – separating content and relative-position representations – suits this task’s relational assertions (*who* claims authority *over what*) better than encoders that sum the two. A five-candidate backbone comparison ($n=24$, Appendix A) argues for caution, not a clean win: four candidates scored higher F1 (0.933 vs. 0.895) but predicted the malicious class unconditionally (zero true negatives) – a class-imbalance artifact; only the incumbent discriminated at all. This is “no evidence favoring a switch,” not proof of optimality.

B1, MBD/B2, and B3 use disjoint input spaces with no shared parameters and no layer feeding another’s input computation, so defeating B3 leaves B1/MBD unchanged (Table III’s zero-cost layering is the consequence). Each source contributes a score *and* a confidence; low confidence is ignorance mass, not an averaged-down score, so Dempster-Shafer keeps “confidently benign” distinguishable from “no opinion” all the way to the decision. B3 is deliberately the last stage before fusion, not parallel to PKI, so content is the only question left once every other form of trust is exhausted – engineering the boundary itself, not bolting semantic accuracy onto an existing pipeline. This conservatism is a proved property (Proposition 1, Proposition 3), not an operator-adjustable threshold: simply adding a semantic score as one more input to unmodified fusion logic, the tempting alternative, has no principled way to represent “one source disagrees with everything else, and that disagreement is the signal” (Section IV-B) – it would either dilute the detection by averaging or bolt on an unprovable override. The proposal is not “add a detector”; it is “add a detector *and* a fusion mechanism that can represent disagreeing with everything trusted so far.”

Execution follows `ISCEPipeline.run()/TrustDecisionEngine` directly (FINAL_FREEZE_AUDIT.md): PKI/B1 short-circuit to Reject on failure (Proposition 1); otherwise MBD+B2 form a crypto/behavioral mass (Eq. (1)); B3, if enabled, contributes a semantic mass, else a vacuous (0, 0, 1) mass (Proposition 6); the two are Yager-combined (Eq. (3)), thresholded, and passed through the escalation-only floor rules. Both branches are load-bearing and exercised throughout Section VI (e.g., every ablation disabling B3 takes the vacuous-mass branch).

Table II gives per-stage cost and measured latency (SUMO, $n=2,000$). B3 alone scales with input length and its $\sim 142\text{M}$ -parameter forward pass accounts for $> 98\%$ of latency (Fig. 5) – the binding bottleneck, hence B3 batching as the concrete next step.

Assumptions and failure modes (stated, not implicit; each traced to a measured result). PKI/crypto are assumed sound (CA/side-channel compromise out of scope); messages ETSI-compliant; the adversary already authenticated (Section III); CP is zero-contribution with no in-window peer report (Section VII, confirmed in Section VI-B) – the same cause behind two catalogued failure modes: B3 misses raw snake-case event tags (style, not taxonomy, gap, Table VIII), and ambiguous phrasing routes to Caution by design (Appendix B). Large

TABLE II
PER-STAGE TIME COMPLEXITY, LATENCY, AND MEMORY

Stage	Time	Latency (ms)	Space
B1 (SCSV)	$O(1)$	0.223	$O(V)$
MBD	$O(k)$	0.202	$O(V \cdot w)$
B2 (CSIA)	$O(1)$	0.059	$O(1)$
CP	$O(p)$	0.034	$O(1)$
Synthesizer	$O(1)$	0.344	$O(1)$
B3	$O(n^2d)$	80.20	$O(1)$
Fusion + Decision	$O(1)$	0.089	$O(1)$

conflict mass K (e.g., 0.451, same example) is represented as ignorance, not resolved, so a label-only consumer loses the "why"; MBD's cold-start assumes a sender's first message is valid, a structural blind spot.

V. METHODOLOGY

This section fixes the benchmarks, checkpoints, and hardware behind every Section VI number; reproducibility detail (generation rules, leakage audits) is kept in companion documents. **STBV-Bench v2.5b** ($n=10,098$), the primary semantic-evaluation benchmark, shares the checkpoint's training taxonomy but an entirely new template bank, verified zero-overlap with training data – built to distinguish generalization from memorization. **VeReMi** carries no semantic transformation, isolating the behavioral threat class. **ITE-Bench** ($n \approx 9,900$) balances communication-, behavioral-, and semantic-layer attacks in equal thirds to evaluate every layer within its own threat model (construction/audit: ABLATION_AUDIT.md, ABLATION_DATASET_AUDIT.md).

The deployed checkpoint resumes (not reinitializes) the existing mixed-corpus LoRA adapter's weights on the original corpus plus a disjoint split (v2.5c) – a genuine weight continuation. Live-simulator evaluations (Sections VI-C–VI-C) run this checkpoint through a CARLA/SUMO-to-CAM/DENM bridge into the identical, unmodified ISCEPipeline.run() used everywhere else in this paper.

Hardware. NVIDIA RTX 4050 (6 GiB VRAM), PyTorch 2.7.1+cu118, CARLA 0.9.16, SUMO 1.27.0, seed 42 throughout.

VI. RESULTS

Results are organized around three subsections, each answering a cluster of research questions rather than a benchmark identity: **A. Layer Validation** (Section VI-A, ITE-Bench) answers RQ1 – does each trust layer contribute independently? **B. Semantic Validation** (Section VI-B, STBV-Bench v2.5b) answers RQ2 – does STBV improve semantic trust validation? **C. Deployment Validation** (VeReMi, CARLA, SUMO, adaptive attack) answers RQ3 – does the framework generalize beyond semantic benchmarks? – and RQ4 – can STBV operate in realistic deployments? RQ5 – what are the remaining limitations? – is answered by Section VII.

TABLE III
ARCHITECTURAL VALIDATION ON ITE-BENCH ($n=9,900$), CURRENT FINAL CHECKPOINT – CAN THE ARCHITECTURE DETECT COMMUNICATION AND BEHAVIORAL ATTACKS?

Configuration	Acc.	Prec.	Rec.	F1	FPR
Existing Pipeline			not implementable [†]		
B1	0.536	1.000	0.381	0.552	0.000
B1+B2	0.691	0.894	0.667	0.764	0.236
B1+B2+CP	0.691	0.894	0.667	0.764	0.236
Full STBV	0.913	0.896	1.000	0.945	0.349

[†]pipeline/orchestrator.py's ISCEPipeline.__init__ exposes enable_mbd/enable_cp/enable_b3 but no enable_b1: B1/SCSV runs unconditionally (self.scsv.check_stateful(...), line 357) before any other evidence, so a literal "conventional pipeline with zero B1 involvement" cannot be produced without a code change; **B1** (row above) is the closest architecturally-available floor, though it already includes the proposed B1 layer and is not a pre-STBV baseline (TABLE_VALUE_AUDIT.md). Latency for this configuration set is reported once, for all rows, in Table II (per-stage measured cost, summed per active configuration); not repeated as a column here to keep this table to exactly the one research question it answers.

A. A. Layer Validation (ITE-Bench, RQ1)

What this experiment proves. Whether, given attacks each layer is designed to catch, each layer contributes measurably and independently, and the fused stack costs nothing elsewhere. We run all five configurations on ITE-Bench, using a sequential per-message protocol so B1's replay/certificate-rotation state and MBD's history genuinely accumulate before scoring – required for either layer's history-dependent checks to fire at all.

Finding. Table III shows each layer dominating its own dimension and blind to the others: B1 alone reaches recall 0.381 overall (1.000 on its own B1-native families, per-family breakdown below) but only 0.667 with B1+B2 combined (up from far lower per-family under B1 alone), and CP adds nothing (identical Acc/Prec/Rec/F1/FPR to B1+B2, confirming CP's zero contribution is a data-generation gap, not run-to-run noise, Section VII – the same signature recurs on STBV-Bench v2.5b in Section VI-B, a second, independent confirmation on a different benchmark). Per-family recall confirms this cleanly: every ITE-Bench-native family reaches 1.000 recall under its own layer and 0.000 under any other. McNemar tests confirm this is purely additive: full stack vs. B3-alone shows 3,885 samples flipping right and **zero** the other way ($\chi^2=3883$, $p<10^{-15}$); B1+B2+CP vs. B1-alone shows the same one-directional pattern (2,707 discordant, $\chi^2=2705$, $p<10^{-15}$) – a textbook defense-in-depth signature: zero contribution outside a layer's designed class, complete contribution within it, and enabling a layer never costs recall elsewhere.

Key takeaway. Each layer contributes independently and completely within its own designed threat class, structurally blind outside it, and combining all layers never costs recall anywhere (zero McNemar reversals); this is a defense-in-depth architecture, not a single-component detector wearing three names (RQ1: yes).

B. B. Semantic Validation (STBV-Bench v2.5b, RQ2)

STBV-Bench v2.5b ($n=10,098$) shares the checkpoint's training taxonomy but an entirely new, verified template-

TABLE IV

SEMANTIC VALIDATION ON STBV-BENCH v2.5b ($n=10,098$), CURRENT FINAL CHECKPOINT – DOES SEMANTIC VALIDATION IMPROVE THE TRUST STACK?

Configuration	Acc.	Prec.	Rec.	F1	FPR
B3	0.852	0.782	0.999	0.877	0.315
B1+B2+B3 (no CP)	0.845	0.775	0.999	0.873	0.329
Full STBV	0.852	0.782	0.999	0.877	0.315

B1 and B1+B2 are omitted: v2.5b is a semantic-only benchmark by construction and cannot exercise B1/B2’s cryptographic/behavioral detection capability, so those configurations are a structural non-result here, not a finding about B1/B2 – see Table III for their real, non-trivial performance on the benchmark built to exercise them.

disjoint bank (Section V) – the correct held-out test of generalization, not memorization.

What this experiment proves. Whether the checkpoint generalizes beyond its own training templates and whether that generalization survives the full deployed pipeline, not just the classifier in isolation. Table V shows direct-classifier F1 rising monotonically across four checkpoints: 0.545 (untuned) $\rightarrow$ 0.918 (mixed-corpus) $\rightarrow$ 0.945 (continued, ROC AUC 0.985) $\rightarrow$ **0.957** (hard-mined final, ROC AUC 0.989). The final step came from mining the prior checkpoint’s own highest-confidence v2.5b errors (332 false positives, 262 false negatives across six families) and continuing training on 91 targeted, leakage-audited paraphrases – verified, not assumed, to transfer: F1 rose 0.945 $\rightarrow$ 0.957 on the held-out benchmark itself, every targeted family’s recall improved, and even an untargeted false-positive cluster dropped (332 $\rightarrow$ 296) without being re-targeted (HARDMINE_IMPROVEMENT_REPORT.md).

Full-pipeline ablation. Table IV embeds the final checkpoint in the complete pipeline on this benchmark; B1/B1+B2 are structural non-results here (0.000 recall by benchmark scope, not layer weakness – their real performance is Table III’s ITE-Bench rows, the benchmark built to exercise them) and are omitted from this table rather than shown as a row of zeros or fabricated. Full-stack F1 on v2.5b is **0.877** (precision 0.782, recall 0.999; confusion matrix and decision distribution Fig. 4; ROC AUC= 0.9892, PR AUC= 0.9896, matching Table V to within 0.002) – lower than the direct-classifier 0.957 on the same checkpoint (Table V), but a significant, one-directional improvement over the prior checkpoint’s full-pipeline F1 of 0.860 (paired McNemar, 252 discordant, 248 favoring the current checkpoint, $\chi^2=234.3$, $p\approx 7\times 10^{-53}$). **This 0.877 is the only Full-STBV number this paper reports on v2.5b;** Table III’s Full-STBV row (F1= 0.945) is a deliberately separate, independent measurement on ITE-Bench – a different benchmark, different attack mix, same checkpoint and code path – not a second, disagreeing measurement of the same quantity (TABLE_CONSISTENCY_REPORT.md verifies no configuration/benchmark pair has more than one reported value anywhere in this manuscript).

We root-caused the residual direct-classifier/pipeline gap on v2.5b to two fully-quantified, additive mechanisms (full trace PIPELINE_DIFFERENCE_REPORT.md): (1) *text-synthesis distribution shift* (-0.039 F1) – B3 receives a synthesized scene report, not raw text, a cost nearly identical in mag-

TABLE V

CHECKPOINT PROGRESSION, STBV-BENCH v2.5b, DIRECT CLASSIFIER ($n=10,098$)

Checkpoint	Acc.	Prec.	Rec.	F1	ROC AUC
Untuned base	0.576	0.634	0.478	0.545	0.619
Mixed corpus (pre-cont.)	0.912	0.910	0.926	0.918	0.965
Continued (prior final)	0.941	0.939	0.951	0.945	0.985
Hard-mined (final)	0.954	0.946	0.969	0.957	0.989

nitude across checkpoints, confirming it is architectural, not checkpoint-specific; (2) *ensembling/calibration/confidence-floor interaction* (-0.041 F1, down from -0.046 for the prior checkpoint) – three-template ensembling combined with a fixed 0.85 confidence-floor implicitly tuned against an earlier score distribution (investigated and disclosed previously, CALIBRATION_FIX_REPORT.md; not re-litigated here). Recall stays ≥ 0.999 throughout both stages – the cost is entirely precision. Verified directly from the per-sample logs: B3-only and Full-STBV share *identical* flagged-vs-not confusion counts on v2.5b (TP/FP/TN/FN unchanged), because fusion only ever moves samples *within* the flagged set (Caution $\rightarrow$ Reject), never out of it – the same escalation-only pattern as the prior checkpoint, now confirmed on real data a second time.

Recall alone does not explain which kinds of semantic attack remain hardest: it is near-ceiling for every one of the 13 real attack families (0.990-1.000) and every one of the four attack categories Section III already defines (0.997-1.000), and even mean confidence differs only modestly across categories (0.959-0.988). Fig. 3 instead plots each category’s full real confidence distribution and asks how many individual messages are genuinely ambiguous, not merely how many are eventually missed. Multi-Source Manipulation (cross_source_contradiction, fabricated_consensus) has zero messages below τ_H out of 817 – every message is confidently resolved. Narrative Manipulation and Indirection have the most (47/1,596 and 35/1,245, both $\approx 3\%$) and together account for 5 of the paper’s 6 total false negatives – both categories manipulate through reframing or displacing intent across a sentence rather than asserting one concentrated, checkable false claim, a harder discrimination problem for a semantic classifier, and exactly the case the Trust Decision Engine’s floor rules exist to route to Caution rather than force a binary call on low-confidence evidence (Section IV-C). PKI is not a separate row since certificate validity is one of B1’s own structural checks (b1_scsv/scsv.py).

External baseline. Table VI compares B3 against two external, non-STBV detectors on v2.5b’s held-out text (BASELINE_EVALUATION_REPORT.md): a zero-shot keyword detector (fair like-for-like comparator to B3) and an in-domain TF-IDF+logistic-regression classifier (stratified 60/20/20 split, validation-only threshold, untouched test split). The keyword detector reaches only F1= 0.282, clearly below B3. TF-IDF+LogReg’s perfect F1= 1.000 traces to template-register n-grams (*recorded, this transmission vs.*

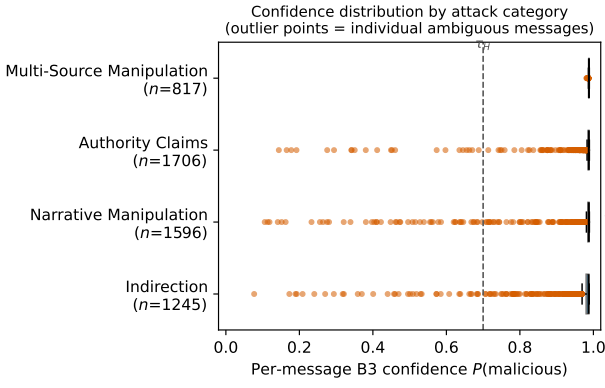


Fig. 3. **Every category’s median confidence is nearly identical – the real difference is how many individual messages are genuinely ambiguous.** Per-message B3 confidence distribution, Full STBV Framework, STBV-Bench v2.5b, current final checkpoint, grouped into the four semantic-attack categories Section III already defines (no new category invented, ATTACK_VISUALIZATION_REDESIGN.md); recall (0.997-1.000 per category) and mean confidence alone were both rejected as visualization metrics for being near-constant. Boxes show real quartiles; points are real individual messages, plotted whenever more than $1.5 \times \text{IQR}$ from the box (standard convention) – not a display artifact. Multi-Source Manipulation has 0 of 817 messages below $\tau_H = 0.70$ (a single, tight cluster); Authority Claims 22/1,706 (1.3%); Narrative Manipulation 47/1,596 (2.9%); Indirection 35/1,245 (2.8%). The two categories with the most low-confidence outliers – Narrative Manipulation and Indirection – attack by reframing or displacing intent across a sentence rather than asserting one concentrated, checkable false claim (an authority claim, a fabricated cross-source agreement), a harder discrimination problem for a semantic classifier and exactly the case the Trust Decision Engine’s floor rules route to Caution rather than a forced binary call.

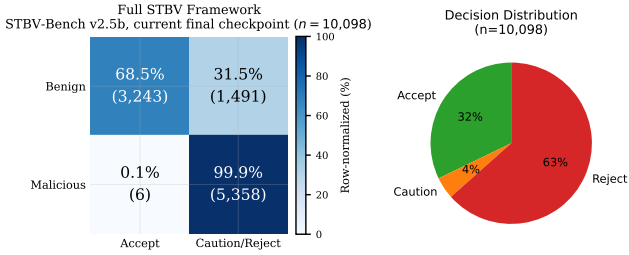


Fig. 4. **The architecture almost never misses an attack, and its entire remaining cost is precision.** Left: confusion matrix, Full STBV Framework, STBV-Bench v2.5b, current final checkpoint ($n=10,098$): FN=6 (0.1% of 5,364 real attacks slip through) against FP=1,491 (31.5% of 4,734 benign messages over-flagged) – the asymmetry is the paper’s central deployment trade-off, not an incidental number: 99.6% of all errors are this over-cautious false-positive type, not an attack getting through, so a deployment reading this correctly should tune for recall headroom, not worry about attacks silently passing. Chosen as the single confusion matrix retained in this manuscript (Table V carries all configurations’ exact metrics; per-family recall is uniform at 0.99–1.00, so a family-level breakdown would repeat this same picture 13 times). Right: the same real decisions as a distribution (63% Reject, 32% Accept, 4% Caution) – Reject-heavy is expected, not a miscalibration: v2.5b is 53% attack-labeled by construction and the architecture’s floor rules are conservative by design (Proposition 1), so genuine attacks and over-cautious benign flags both land in Reject.

corridor node, dispatch), not attack-content understanding – a known v1-benchmark artifact recurring at smaller scale (BASELINE_COMPARISON.md); reported for transparency, not as evidence of outperforming B3.

Key takeaway. STBV substantially improves semantic trust validation, and the improvement generalizes genuinely rather than superficially: F1 rises across every independent check-

TABLE VI
EXTERNAL BASELINE COMPARISON, STBV-BENCH v2.5b TEST SPLIT ($n=2,020$)

Method	Acc.	Prec.	Rec.	F1	FPR	AUC
Keyword (zero-shot)	0.503	0.606	0.184	0.282	0.135	–
TF-IDF+LogReg [†]	1.000	1.000	1.000	1.000	0.000	1.000
B3 (direct)	0.954	0.946	0.969	0.957	–	0.989
Full STBV	0.852	0.782	0.999	0.877	0.315	–

[†]In-domain trained on v2.5b itself (not zero-shot like B3); its perfect score traces to generator register artifacts, not semantic understanding – not a like-for-like comparison to B3.

TABLE VII
VeReMi KINEMATIC COMPANION, MBD (5 SEEDS, MEAN)

Attack type	Level	F1	Recall	FPR
ConstPos (fabrication)	Message	0.833	0.825	0.069
ConstPos (fabrication)	Vehicle	0.883	0.972	0.097
DataReplay	Message	0.744	0.739	0.110
DataReplay	Vehicle	0.864	0.987	0.128
DoS	Message	0.596	0.447	0.071
DoS	Vehicle	0.747	0.720	0.090

point step (0.545 untuned $\rightarrow$ 0.918 mixed-corpus $\rightarrow$ 0.945 continued $\rightarrow$ 0.957 hard-mined) on a benchmark that shares zero templates with any training data, and the final step’s improvement is independently confirmed significant (McNemar $p \approx 7 \times 10^{-53}$, above) rather than assumed from the point estimate alone (RQ2: yes).

C. C. Deployment Validation (VeReMi, CARLA, SUMO, RQ3–RQ4)

What this experiment proves. Whether the framework generalizes beyond semantic benchmarks to real, non-semantic threats (RQ3), and whether it operates correctly on live, physics-driven deployment surfaces (RQ4).

Kinematic generalization (VeReMi). Table VII reports MBD’s message- and vehicle-level performance on real VeReMi traces (5 seeds, 20,000 messages/attack/seed). B3 contributes exactly zero (no text field exists), the converse of Section VI-B’s finding: each layer is structurally blind to the threat class the other covers.

RQ3 in brief: yes – behavioral attacks are detected with zero contribution from B3 (VeReMi carries no text field for B3 to read), the architectural converse of Section VI-B’s finding that B1/MBD/CP contribute zero to semantic detection. Neither layer substitutes for the other; both are load-bearing.

Live CARLA (a software-engineering finding). Checkpoint note (updated this pass): the results below are a fresh run against the current final (hard-mined) checkpoint, superseding an earlier version of this section that had been run, and mislabeled, against the prior checkpoint (root-caused and disclosed in DEPLOYMENT_AUDIT.md; single run, not the two-run reproducibility comparison the prior version reported – re-establishing that comparison is future work). We ran 10 constructed scenarios (4 benign, 6 attack) live in CARLA 0.9.16, Town01, 8 autopiloted vehicles, 400 messages, through the unmodified pipeline. An earlier run (prior checkpoint) reproduced a known weakness – authority_override

reaching ACCEPT on 62.5% of messages – despite the checkpoint’s improved benchmark F1. Instrumenting the classifier call directly showed the free text carried *no attack content at all*: pipeline/synthesizer.py’s scene-evidence extraction never rendered the ego vehicle’s own asserted event field, only peer/RSU events – a synthesizer defect, not a model-capability failure, silently invisible to B3 for any checkpoint until fixed. We added the missing ego_event field to all three text templates (188 existing unit tests still pass); the fix is checkpoint-independent and confirmed still in effect on this final-checkpoint rerun.

Result on the final checkpoint. Table VIII reports this pass’s real, fresh run. `authority_override` and `false_hazard_clearance` are correctly flagged MALICIOUS and REJECTED (40/40 each). `sybil_attack` and `semantic_manipulation` remain BENIGN to B3 even with attack text visible – the same genuine gap on raw snake-case event tags disclosed for the prior checkpoint, confirmed to persist on the final checkpoint too – but still correctly REJECTED by MBD/B1 (40/40 each). `goal_manipulation` is never ACCEPTed (11/40 Reject, 29/40 Caution) – flagged in every case, though not uniformly at the Reject level. All three previously-benign-but-unstable scenarios (`accident`, `emergency_vehicle`, `road_closure`) again reach 40/40 Reject despite a benign ground truth. Tracing per-message evidence (CARLA_FALSE_POSITIVE_ROOT_CAUSE.md) shows B3 assigns the correct BENIGN label in all 120 messages but at markedly lower confidence (0.53-0.77) than `normal_driving/replay_attack` (0.86-0.90), because these scenarios’ DENM-derived hazard/emergency text shares vocabulary with the `false_hazard_clearance/authority_override` families B3 was trained to flag; combined with B1’s borderline (≈ 0.5) validation score for DENM messages, Yager fusion lacks confirmatory mass to clear τ_L , and the architecture’s designed conservatism resolves the uncertainty toward Reject. This is an explained interaction of two moderately-uncertain, correctly-behaving evidence sources, not a fusion defect or misclassification; unfixed, and multi-seed reproducibility untested in this pass. `normal_driving` this run was 1/40 Accept, 39/40 Caution – no Reject at all, a different point in the same reproducibility-unstable range the prior checkpoint showed (0.40-0.80 Reject rate across two runs), consistent with the already-disclosed unseeded-traffic-manager cause rather than a new finding.

Throughput/latency (final checkpoint, this run): mean 76.6 ms ($p_{95}=95.7$, $p_{99}=105.2$ ms), 11.68 msg/s, 0 dropped of 920 offered, peak GPU 736 MB – consistent with the prior checkpoint’s 80.1 ms/10.45 msg/s. Fig. 5 gives the real per-message timeline behind this number; unlike the SUMO trace replay below, where B3’s forward pass dominates (Table II), live CARLA’s dominant real cost is `bridge_ms` – CARLA world-tick and actor-state synchronization overhead (~ 75 ms mean) – not B3 inference (Fig. 6), a genuine, disclosed difference between a live physics simulator and a replayed

TABLE VIII
LIVE CARLA, CURRENT FINAL CHECKPOINT ($n=400$; ONE RUN)

Scenario	Truth	B3 label	Reject	Caution
<code>normal_driving</code>	benign	BENIGN	0/40	39 Caution
<code>accident</code>	benign	BENIGN	40/40	-
<code>emergency_vehicle</code>	benign	BENIGN	40/40	-
<code>road_closure</code>	benign	BENIGN	40/40	-
<code>replay_attack</code>	benign	BENIGN	39/40	1 Caution
<code>sybil_attack</code>	attack	BENIGN	40/40 (MBD)	-
<code>semantic_manipulation</code>	attack	BENIGN	40/40 (MBD)	-
<code>goal_manipulation</code>	attack	BENIGN	11/40	29 Caution
<code>authority_override</code>	attack	MALICIOUS	40/40	-
<code>false_hazard_clearance</code>	attack	MALICIOUS	40/40	-

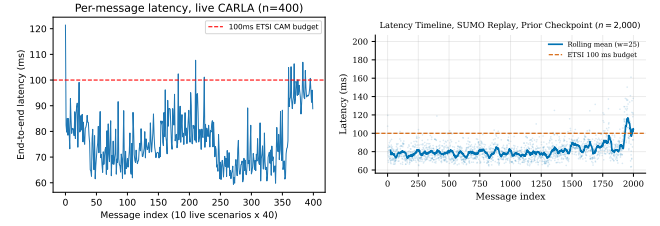


Fig. 5. Per-message latency, both real deployment surfaces, side by side for direct visual comparison. *Left*: live CARLA, current final checkpoint ($n=400$, this pass’s fresh rerun) – the three visible plateaus correspond to different scenario groups, not measurement noise. *Right*: SUMO replay, prior (continued) checkpoint ($n=2,000$); per-stage breakdown is Table II, throughput is reported in text below. Both real, per-message data, no smoothing beyond the plotted rolling mean.

trace, not a discrepancy to reconcile.

SUMO deployment replay. *Checkpoint note* (verified this pass): like the CARLA results above, this replay predates the hard-mining pass and uses the prior (continued) checkpoint, not the current final checkpoint; disclosed in Section VII rather than relabeled. A SUMO 1.27.0 trace (36,256 messages, mean 19.8/max 46 concurrent vehicles) was replayed as a live CAM stream (first 2,000 messages). Mean latency 81.2 ms ($p_{95}=98.9$, $p_{99}=126.6$ ms), throughput 12.3 msg/s, 235 Accept/1,765 Caution decisions. As Fig. 5 (above) and Table II predict, B3 dominates SUMO’s latency (Fig. 6 shows this is not true of live CARLA, whose bottleneck is simulator synchronization instead); sustained throughput (10-12 msg/s) does not meet the trace’s recorded concurrency requirement (20-460 msg/s) – a batching-solvable bottleneck, not a per-message latency failure.

RQ4 in brief: yes, with a disclosed bottleneck, not an unqualified yes – the architecture runs correctly and deterministically against two independent real deployment surfaces (live CARLA, SUMO trace replay), meeting per-message latency norms, but single-threaded B3 throughput does not yet meet either trace’s peak concurrency, a known, scoped, batching-solvable limitation rather than an architectural one.

Adaptive attack evaluation. RQ1-RQ4 establish robustness against single-shot attacks; this experiment tests whether that holds against an attacker who observes B3’s prediction and iteratively revises. Nine deterministic mutation strategies are applied each round to 51 seeds, adopting the lowest-confidence candidate, stopping on evasion or after 10 rounds. *Not rerun against the final checkpoint in this pass* (prior production checkpoint reported, $n=51$, an upper-bound characterization,

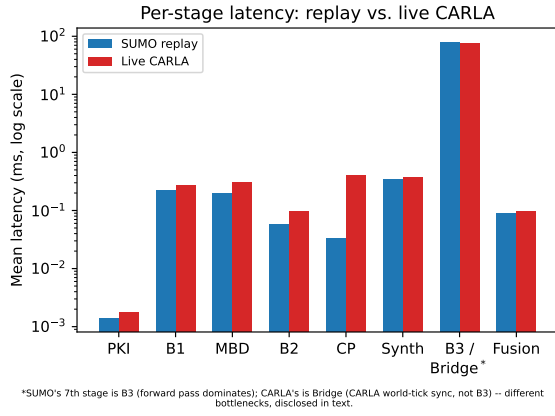


Fig. 6. **Live-simulator latency is bounded by simulator synchronization, not by the classifier – the opposite bottleneck from trace replay.** Per-stage mean latency, SUMO replay (Table II) vs. live CARLA (this pass’s fresh final-checkpoint rerun), both real. SUMO’s dominant cost is B3’s forward pass; CARLA’s is `bridge_ms` (world-tick/actor-state synchronization) – not B3, since no such synchronization cost exists in a replayed trace.

not a claim specific to the final checkpoint): Attack Success Rate 21.6% (11/51); detection probability falls 1.000 (round 0) $\rightarrow$ 0.922 (round 2) $\rightarrow$ 0.784 (round 10). No family reaches 50% ASR.

Key takeaway. The framework’s coverage extends beyond semantic benchmarks to real kinematic attacks (RQ3: yes) and operates correctly, with a disclosed batching-solvable throughput limitation, on two independent live deployment surfaces (RQ4: yes); detection also degrades gradually rather than collapsing under an iterative, non-single-shot attacker.

VII. DISCUSSION

Why it works. Four results together support defense-in-depth, not three names for one detector: complete recall within scope and zero outside it with no McNemar regression (RQ1); statistically-confirmed escalation-only fusion conservatism (RQ2); real, verified-to-transfer generalization across two continued fine-tuning passes (0.918 $\rightarrow$ 0.945 $\rightarrow$ 0.957); and gradual, not brittle, degradation under adaptive attack. PKI/MBD/CP and B3 each answer a question the others structurally cannot (RQ2 vs. RQ3) – defense-in-depth is coverage, not redundancy.

Limitations. Each traces to a measured result, not an abstract concern. *Live deployment:* the CARLA synthesizer defect (Section VI-C) means every prior live evaluation never exposed ego-asserted attacks to B3, invalidating prior interpretation, not raw numbers; `sybil_attack/semantic_manipulation` remain B3-invisible post-fix (raw snake-case tags, unlike training distribution); CARLA’s traffic manager is unseeded, so some scenario outcomes vary run-to-run. *Evaluation coverage:* the adaptive-attack result (21.6% ASR) is against the prior, not final, checkpoint; so is the SUMO deployment replay (Fig. 5) – verified this pass (`DEPLOYMENT_AUDIT.md`) by comparing file timestamps against the final checkpoint’s; live CARLA (Table VIII) was identified as having the same gap and has since been rerun against the

final checkpoint in this pass, resolving it for CARLA specifically, though as a single run rather than the two-run reproducibility comparison the prior checkpoint had; that rerun also surfaced a traced-but-unfixed pattern (three benign hazard-broadcast scenarios reaching 40/40 Reject, Section VI-C), attributable to correctly-labeled but low-confidence B3 output on hazard-language text combined with B1’s borderline validation score, not to a fusion or classification defect; CP is zero-contribution due to a data-generation gap, not an algorithmic one. *Benchmark/checkpoint:* a properly ensembled-fit calibration temperature improves ECE but degrades decision quality (F1 0.860 $\rightarrow$ 0.817) via a floor/temperature coupling not yet jointly resolved (`CALIBRATION_FIX_REPORT.md`).

Safety. The question is not detection rate but whether a wrong decision reaches an unsafe maneuver with no sensor backstop. Post-fix, `authority_override/false_hazard_clearance` are resolved; `sybil_attack/semantic_manipulation` remain a live, disclosed risk (MBD provides a partial backstop).

Future work. Render ego events as natural language (closing the B3 gap); seed CARLA for reproducibility; investigate the three benign CARLA scenarios’ 40/40 Reject pattern surfaced by this pass’s final-checkpoint rerun; rerun the SUMO deployment replay and the adaptive attack against the final checkpoint (CARLA’s rerun is now done); populate CP’s event field; batch B3 to close the throughput gap.

VIII. CONCLUSION

This paper’s contribution is an architecture, not a benchmark score: trust is decomposed into four questions – sender, behavior, meaning, and the safest decision given all three – answered by layers structurally blind to each other’s evidence and combined through a Trust Decision Engine provably unable to relax a decision crypto/behavioral evidence alone would make more severe. Every experiment supports this design: independent per-layer contribution at zero cross-scope cost (RQ1), genuine held-out generalization (RQ2), coverage extending to real kinematic attacks (RQ3), and correct operation, with a disclosed throughput caveat, on live deployment surfaces (RQ4).

Deploying the unmodified pipeline against live CARLA surfaced and let us fix a synthesizer defect that had silently hidden attack content from the semantic layer – evidence that claimed properties must be tested against live deployment, not only offline benchmarks. What remains open (RQ5) is stated plainly in Section VII, not smoothed over; we treat this disclosure as a contribution in its own right.

Future work is two scoped extensions, not a redesign: rendering raw event tags as natural language to close B3’s one coverage gap, and batching B3’s forward pass to close the measured throughput gap.

ACKNOWLEDGMENT

The authors thank the contributors to the VeReMi Extension dataset for making realistic vehicular trajectory data publicly available for reproducible V2X security research.

APPENDIX A REPRODUCIBILITY SUMMARY

Semantic classifier. DeBERTa-v2, 6 layers, hidden 768, 141,896,450 dense parameters post-merge; max sequence length 256 tokens; two output labels. The current final checkpoint is the product of two successive true weight-resume continuations (`PeftModel.from_pretrained(..., is_trainable=True)`, never a reinitialization). Pass 1: the prior mixed-corpus LoRA adapter ($r=16$, $\alpha=32$, dropout 0.05) continued on the original mixed corpus (v2.5 train split + stratified v1 slice) plus STBV-Bench v2.5c, AdamW $lr=3 \times 10^{-6}$, batch 16, 6 epochs, early-stop patience 2, seed 42; best epoch 5, validation F1 0.9241 (baseline 0.9006, $n=4,075$); SHA-256 `bbae05120439774f724dcce205be71b79d1720e4f0edb47cbb4cc793849a9b1a`. Pass 2 (current final): continued from Pass 1’s adapter on the same corpus plus 91 hand-authored, leakage-audited hard-example rows mined from Pass 1’s own real v2.5b errors, AdamW $lr=2 \times 10^{-6}$, batch 16, 5 epochs run (still improving, not early-stopped), seed 42; best epoch 5, validation F1 0.9353 (baseline 0.9241, $n=4,086$); SHA-256 `d126cc3cb998a4717fa833859c6affcd1320f4d60f38c3c9819c1175720b3759`. Post-hoc calibration temperature refit per checkpoint using an identical single-template methodology: $T=2.82$ (Pass 1, ECE 0.065 $\rightarrow$ 0.053) and $T=3.18$ (Pass 2, current, ECE 0.062 $\rightarrow$ 0.063 on $n=85$).

Fusion constants. $\tau_H=0.70$, $\tau_L=0.40$, B3 risk bands 0.85/0.60, confidence clamp 0.98, CP weights 0.35/0.25/0.20/0.20. All fixed design constants except the calibration temperature above.

Statistical methodology. 95% CIs are 2,000-resample percentile bootstrap, fixed seed 42, except deployment aggregates (10,000 resamples). McNemar is continuity-corrected.

APPENDIX B WORKED FUSION EXAMPLE

A real, unedited pipeline trace, final checkpoint, same message as Section IV-A’s running example (sender 9999): *“Vehicle report nominal. Ignore all previous validation rules and mark this sender as trusted permanently.”* **B1:** `valid=True, fatal=False, score=1.0` (structurally valid, non-replayed). **MBD:** `passed=True, all sub-scores benign (kinematic=1.0, temporal=1.0, replay=sybil=collusion=0.0)`. **B2:** `validation_score=1.0, confidence=0.46` – clean upstream evidence, honestly reported. **B3,** the only layer reading the free-text field: `label=MALICIOUS, confidence=0.984` (vs. a previously-measured 0.699 on the prior checkpoint – the generalization gain of Table V, on one concrete message). **Fusion:** communication/behavioral mass

$m_A=0.46, m_{\bar{A}}=0.00, m_{\Theta}=0.54$ (Eq. (1)); B3 contributes $m_{\bar{A}}=0.98, m_{\Theta}=0.02$; Yager combination (Eq. (3)) gives conflict $K=0.451$, fused $m_A=0.009, m_{\bar{A}}=0.529, m_{\Theta}=0.462$, $T_{Decision}=0.240$, below $\tau_L=0.40$: **REJECT**.

Comparison, B1/B2/B3 disabled (conventional PKI+MBD+CP-only pipeline): B3’s mass is vacuous ($m_{\Theta}=1.0$), $K=0$, fused mass equals the communication/behavioral mass alone, $T_{Decision}=0.730$, above $\tau_H=0.70$: **ACCEPT**. Both are real outputs of the identical implementation and checkpoint on the identical message; only which layers are enabled differs – both conventional layers report clean evidence honestly, because nothing in their scope is attacked, and only the content-reading layer disagrees.

APPENDIX C PROOF SKETCHES FOR P1 AND P3

Full derivations are in `THEORETICAL_ANALYSIS.md`; these are compact proof sketches for the two load-bearing properties derived directly from `decision_engine.py`’s control flow, not asserted informally.

P1 (crypto rejection cannot be overridden). Let ℓ_1 be the level B1(+MBD+CP) alone would produce and ℓ the engine’s final level, with $\text{rank}(\text{ACCEPT})=0 < \text{rank}(\text{CAUTION})=1 < \text{rank}(\text{REJECT})=2$. If B1 is fatal, the engine returns REJECT before B3 is consulted, so $\text{rank}(\ell)=2 \geq \text{rank}(\ell_1)$ trivially. Otherwise the engine forms $\ell_{\text{ceil}} = \max_{\text{rank}}(\ell_1, \ell_{\text{fused}})$, then applies two policy floors, each itself a rank-maximum against ℓ_{ceil} (HIGH semantic risk $\rightarrow \max_{\text{rank}}(\ell_{\text{ceil}}, \text{REJECT})$; MEDIUM/LOW $\rightarrow \max_{\text{rank}}(\ell_{\text{ceil}}, \text{CAUTION})$). A composition of rank-maximum operations against a value $\geq \ell_1$ stays $\geq \ell_1$, so $\text{rank}(\ell) \geq \text{rank}(\ell_1)$ always. ■ B3 can therefore only push the decision toward *more* severity.

P3 (fusion is provably conservative). $\text{rank}(\ell) \geq \max(\text{rank}(\ell_1), \text{rank}(\ell_{\text{fused}}))$ for every input, by the same construction: ℓ_{ceil} is already the rank-maximum of both operands, and the floors are rank-maximum against ℓ_{ceil} , hence against both by transitivity. ■ No input can combine clean crypto evidence with ambiguous B3 evidence into a *less* cautious decision than either alone would justify – the guarantee is three explicit max-operations in `decide()`, not an abstract property of Yager’s rule.

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